

Multivalued Mappings & Fixed Points

Chapter 5.4: Fixed Point Theorems for Multifunctions

Based on Pathak's Introduction to Nonlinear Analysis

Fixed Point Theory Course

July 14, 2026

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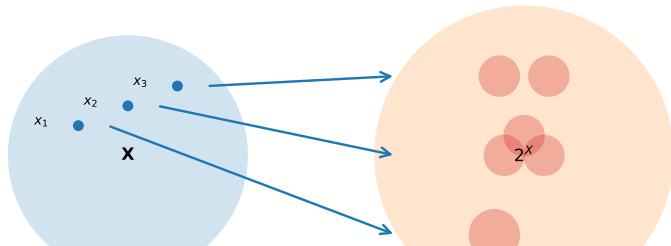
What is a Multivalued Mapping?

Definition

A **multivalued mapping** (or **multifunction**) is a map

$$T : X \rightarrow 2^X$$

that assigns to each point $x \in X$ a **set** $T(x) \subseteq X$ rather than a single point.



Notation and Basic Concepts

- **Domain:** $D(T) = \{x \in X : T(x) \neq \emptyset\}$
- **Range:** $R(T) = \bigcup_{x \in D(T)} T(x)$
- **Graph:** $\text{Gr}(T) = \{(x, y) : y \in T(x)\}$
- **Inverse:** $T^{-1}(B) = \{x \in X : T(x) \cap B \neq \emptyset\}$

Example

Let $X = \mathbb{R}$ and define $T(x) = [x - 1, x + 1]$. Then:

- $D(T) = \mathbb{R}$ (entire domain)
- $T(0) = [-1, 1]$, $T(2) = [1, 3]$
- Each point maps to an interval

Fixed Points for Multivalued Mappings

Definition

A point $x_0 \in X$ is a **fixed point** of the multivalued mapping $T : X \rightarrow 2^X$ if

$$x_0 \in T(x_0)$$

Key Difference from Single-Valued Case:

- For single-valued $f : X \rightarrow X$: fixed point satisfies $f(x_0) = x_0$
- For multivalued $T : X \rightarrow 2^X$: fixed point satisfies $x_0 \in T(x_0)$
- Set membership instead of equality

Example

For $T(x) = [x - 1, x + 1]$:

- Is $x = 0$ a fixed point? Need $0 \in [-1, 1]$. ✓Yes!
- Is $x = 3$ a fixed point? Need $3 \in [2, 4]$. ✓Yes!
- Is $x = 10$ a fixed point? Need $10 \in [9, 11]$. ✓Yes!

Historical Development

- **1941:** Kakutani proved fixed point theorem for multivalued mappings in $\mathbb{R}^n$
- **1950:** Bohnenblust & Karlin extended to infinite-dimensional Banach spaces
- **1952:** Fan generalized result to locally convex spaces
- **1969:** Nadler Jr. developed theory of multivalued contractions with Hausdorff metric
- **1970s:** Assad, Kirk, Markin, Smithson advanced theory further

Applications:

- Control theory and differential equations
- Game theory and economics
- Variational inequalities
- Differential inclusions

Measuring Distance Between Sets

To study contractive multivalued mappings, we need a metric on **sets**, not just points.

Definition

Let (X, d) be a metric space. For non-empty sets $A, B \subseteq X$, define:

$$d(a, B) = \inf_{b \in B} d(a, b) \quad (\text{distance from point to set})$$

$$\rho(A, B) = \sup_{a \in A} d(a, B) \quad (\text{directed Hausdorff distance})$$

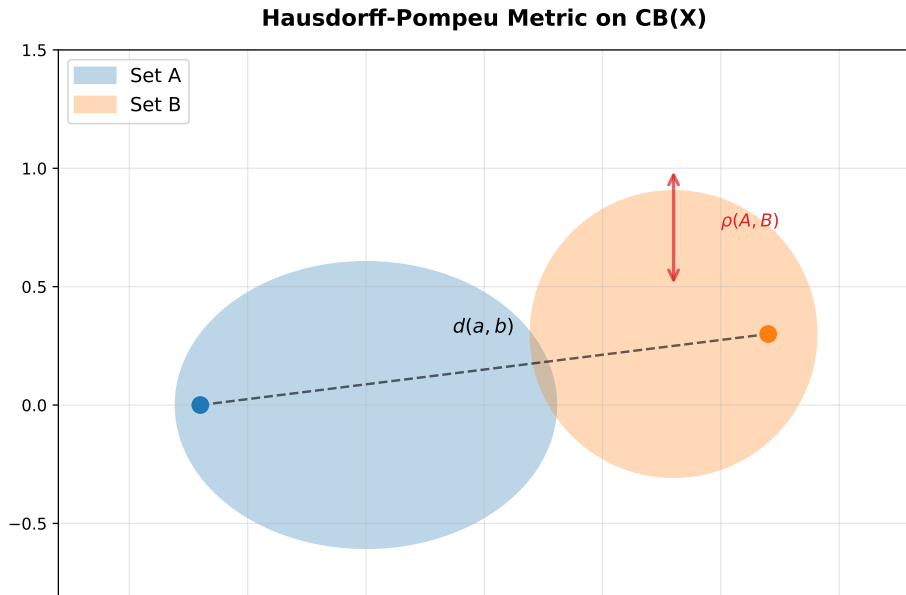
$$H(A, B) = \max\{\rho(A, B), \rho(B, A)\} \quad (\text{Hausdorff metric})$$

Alternative Definition

$$H(A, B) = \inf\{\varepsilon > 0 : A \subseteq N(\varepsilon, B) \text{ and } B \subseteq N(\varepsilon, A)\}$$

where $N(\varepsilon, C) = \{x \in X : d(x, c) < \varepsilon \text{ for some } c \in C\}$

Hausdorff-Pompeu Metric Visualization



Properties of the Hausdorff-Pompeu Metric

Theorem

Let (X, d) be a complete metric space. Then:

- ① $(CB(X), H)$ is a complete metric space, where $CB(X)$ is the set of nonempty closed bounded subsets
- ② $K(X)$ (nonempty compact subsets) is a closed subspace of $CB(X)$
- ③ For single points: $H(\{a\}, \{b\}) = d(a, b)$

Key Observations

- The Hausdorff metric depends on the underlying metric d on X
- Two equivalent metrics on X may not generate equivalent Hausdorff metrics
- The completeness of (X, d) ensures completeness of $(CB(X), H)$

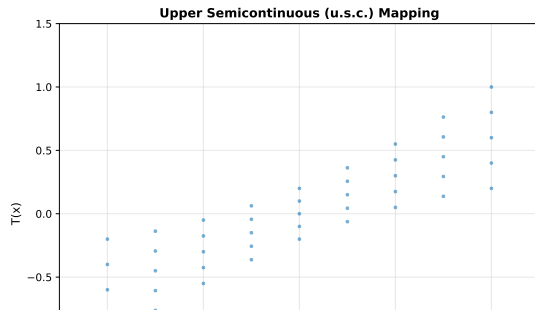
Upper Semicontinuity

Definition

A multifunction $T : X \rightarrow 2^Y$ is **upper semicontinuous (u.s.c.)** if:

For each closed set $B \subseteq Y$, $T^{-1}(B) = \{x : T(x) \cap B \neq \emptyset\}$ is closed

Equivalent characterization: For each $x \in X$ and open set $U \supseteq T(x)$, there exists an open neighborhood V of x such that $U \supseteq T(y)$ for all $y \in V$.



Definition of Upper Semicontinuity

$T: X \rightarrow 2^Y$ is u.s.c. if for each closed $B \subseteq Y$,
the set $T^{-1}(B) = \{x : T(x) \cap B \neq \emptyset\}$ is closed

Equivalently: For each $x \in X$ and open set $U \supseteq T(x)$,
there exists open $V \ni x$ such that $U \supseteq T(y)$ for all $y \in V$

Lower Semicontinuity and Other Notions

Definition

A multifunction $T : X \rightarrow 2^Y$ is **lower semicontinuous (l.s.c.)** if:

For each open set $U \subseteq Y$, $\{x : T(x) \cap U \neq \emptyset\}$ is open

A multifunction is **continuous** if it is both u.s.c. and l.s.c.

Additional Concepts

- **Point closed:** $T(x)$ is closed for all x
- **Point compact:** $T(x)$ is compact for all x
- **Point convex:** $T(x)$ is convex for all x

Remark: Upper semicontinuity is the most natural generalization of continuity for multifunctions.

Fan's Fixed Point Theorem

Theorem (Fan, 1952)

Let X be a normed linear space, K a nonempty, compact and convex subset of X . Let $T : K \rightarrow 2^K$ be a multifunction such that:

- ① *For each $x \in K$: $T(x)$ is nonempty, closed, and convex*
- ② *T is upper semicontinuous*

Then there exists a fixed point $x^ \in K$ such that $x^* \in T(x^*)$.*

Key Features:

- Generalizes Kakutani's theorem from $\mathbb{R}^n$ to normed spaces
- Works in locally convex spaces
- No contraction property needed—only compactness and convexity
- Purely topological argument

Himmelberg's Generalization

Theorem (Himmelberg, 1972)

Let K_1 be a nonempty and convex subset of a normed linear space X . Let $T : K_1 \rightarrow 2^{K_1}$ be a multifunction such that:

- ① T is upper semicontinuous
- ② For each $x \in K_1$: $T(x)$ is closed and convex
- ③ $T(K_1) \subseteq C$ for some compact $C \subseteq K_1$

Then T has a fixed point in K_1 .

Improvement over Fan's Theorem:

- Allows non-compact K_1
- Only requires that $T(K_1)$ be contained in a compact subset
- Proof uses topological degree theory

Min-Max Theorem

Theorem (Consequence of Himmelberg's Theorem)

Let K_1, K_2 be compact subsets of normed spaces, and let $f : K_1 \times K_2 \rightarrow \mathbb{R}$ be continuous. If for any $x_0 \in K_1$ and $y_0 \in K_2$:

- $\{x \in K_1 : f(x, y_0) = \max_{\xi \in K_1} f(\xi, y_0)\}$ is convex
- $\{y \in K_2 : f(x_0, y) = \min_{\eta \in K_2} f(x_0, \eta)\}$ is convex

Then:

$$\max_{x \in K_1} \min_{y \in K_2} f(x, y) = \min_{y \in K_2} \max_{x \in K_1} f(x, y)$$

Application: Game theory—guarantees saddle point existence

Nadler's Fixed Point Theorem

Theorem (Nadler Jr., 1969)

Let (X, d) be a complete metric space and $T : X \rightarrow CB(X)$ a multivalued mapping. If there exists $k \in (0, 1)$ such that:

$$H(T(x), T(y)) \leq k \cdot d(x, y) \quad \text{for all } x, y \in X$$

then T has a fixed point.

Key Properties:

- Generalizes Banach Contraction Principle to multivalued mappings
- Uses Hausdorff metric for set-valued images
- Requires complete metric space
- Contraction constant $k < 1$ is necessary

Comparison with Single-Valued Case

	Single-Valued	Multivalued
Condition	$d(f(x), f(y)) \leq k \cdot d(x, y)$	$H(T(x), T(y)) \leq k \cdot d(x, y)$
Output	Point	Set

Proof Idea of Nadler's Theorem

Construction of Fixed Point Sequence:

- 1 Start with arbitrary $x_0 \in X$
- 2 Choose $x_1 \in T(x_0)$ such that $d(x_0, x_1) \leq \frac{1}{1-k} H(x_0, T(x_0))$
- 3 Inductively: choose $x_{n+1} \in T(x_n)$ such that $d(x_n, x_{n+1}) \leq k^n \cdot d(x_0, x_1)$

Key Steps:

- Show that $\{x_n\}$ is Cauchy:

$$d(x_m, x_n) \leq \frac{k^n}{1-k} d(x_0, x_1) \rightarrow 0 \text{ as } n \rightarrow \infty$$

- Completeness: $x_n \rightarrow p$ for some $p \in X$
- Upper semicontinuity: $p \in T(p)$ (closing operation)

If $H^+(T(x), T(y)) \leq k \cdot H^+(x, y)$ for $k \in (0, 1)$
then T has a unique fixed point

Numerical Example: Nadler's Theorem

Consider $T : [0, 1] \rightarrow CB([0, 1])$ defined by

$$T(x) = \left[\frac{x}{3}, \frac{x}{2} \right]$$

```
import numpy as np

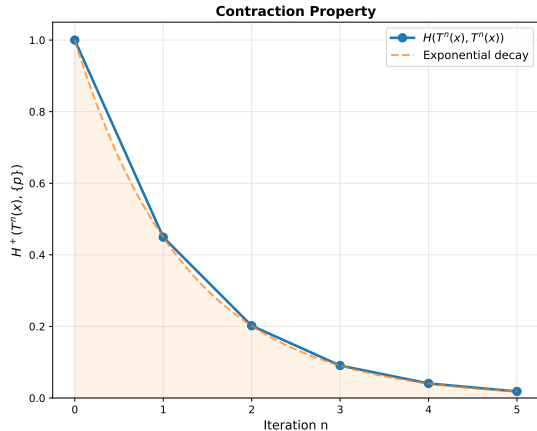
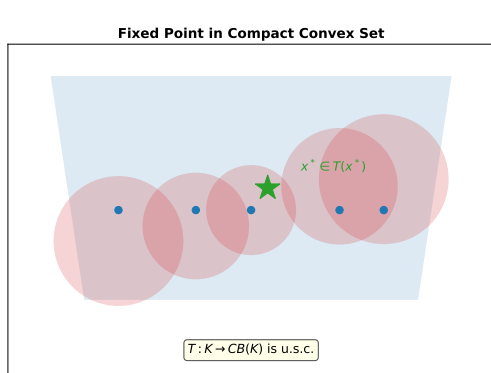
def hausdorff_distance(A, B):
    """Compute Hausdorff distance between intervals"""
    return max(abs(A[0] - B[0]), abs(A[1] - B[1]))

def T(x):
    """Multivalued mapping"""
    return [x/3, x/2]

# Check Nadler condition
x, y = 0.2, 0.5
Tx, Ty = T(x), T(y)
H_dist = hausdorff_distance(Tx, Ty)
k_times_d = 1/3 * abs(x - y) # Contraction rate k = 1/3

print(f"H(T({x}), T({y})) = {H_dist:.4f}")
print(f"(1/3) * d({x}, {y}) = {k_times_d:.4f}")
print(f"Nadler condition satisfied: {H_dist <= k_times_d}")
```

Visualization: Nadler's Contraction



- **Left:** Compact convex set K with multivalued mapping showing nested sets
- **Right:** Exponential convergence of iterates to fixed point

Coincidence Points

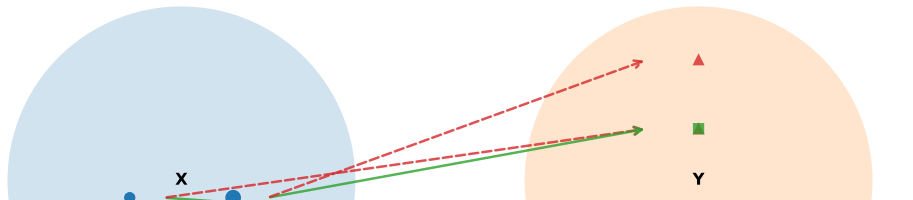
Definition

A point $p \in X$ is a **coincidence point** of mappings $f, g : X \rightarrow Y$ if

$$f(p) = g(p)$$

For multivalued mappings $f, g : X \rightarrow 2^Y$:

Coincidence point: $f(p) \cap g(p) \neq \emptyset$



Fixed Point Theorems for Two Mappings

Theorem (Generalization to Two Mappings)

Let (X, d) be a complete metric space. Let $S, T : X \rightarrow CB(X)$ satisfy the condition:

$$H(S(x), T(y)) \leq k \cdot d(x, y) \quad \text{for all } x, y \in X$$

for some $k \in (0, 1)$. Then there exists $p \in X$ such that $p \in S(p) \cap T(p)$.

Common Fixed Points

If both S and T satisfy:

$$H(S(x), S(y)) \leq k_S \cdot d(x, y), \quad H(T(x), T(y)) \leq k_T \cdot d(x, y)$$

then they have a common fixed point.

Applications: Variational inequalities, nonlinear integral equations, differential inclusions

Hybrid Mappings and Coincidence

Definition

Mappings $f : X \rightarrow X$ and $T : X \rightarrow 2^X$ are said to have a **coincidence point** at p if

$$f(p) \in T(p)$$

Theorem (Coincidence for Hybrid Mappings)

Let $f : X \rightarrow X$ be continuous and $T : X \rightarrow CB(X)$ be u.s.c. in a compact metric space. If $H(T(x), T(y)) \leq k \cdot d(f(x), f(y))$ for $k < 1$, then f and T have a coincidence point.

Key Applications:

- Periodic points of iterative schemes
- Common fixed points of commuting families
- Solutions to differential inclusions

Asymptotic Contraction

Definition

A multivalued mapping $T : X \rightarrow 2^X$ is **asymptotically I-contractive** if there exist sequences $\{k_n\}$, $\{\alpha_n\}$ with:

$$\sum_{n=1}^{\infty} k_n < \infty, \quad \sum_{n=1}^{\infty} \alpha_n < \infty$$

such that for all $x, y \in X$:

$$H(T^n(x), T^n(y)) \leq k_n \cdot d(x, y) + \alpha_n$$

Motivation:

- Relaxes strict contraction requirement
- Allows error terms that decrease overall
- Captures behavior of iterative algorithms with diminishing step sizes

Fixed Point Theorem for Asymptotic Contractions

Theorem

Let (X, d) be a complete metric space and $T : X \rightarrow CB(X)$ asymptotically I -contractive. If $\sum_{n=1}^{\infty} k_n < \infty$ and $\sum_{n=1}^{\infty} \alpha_n < \infty$, then:

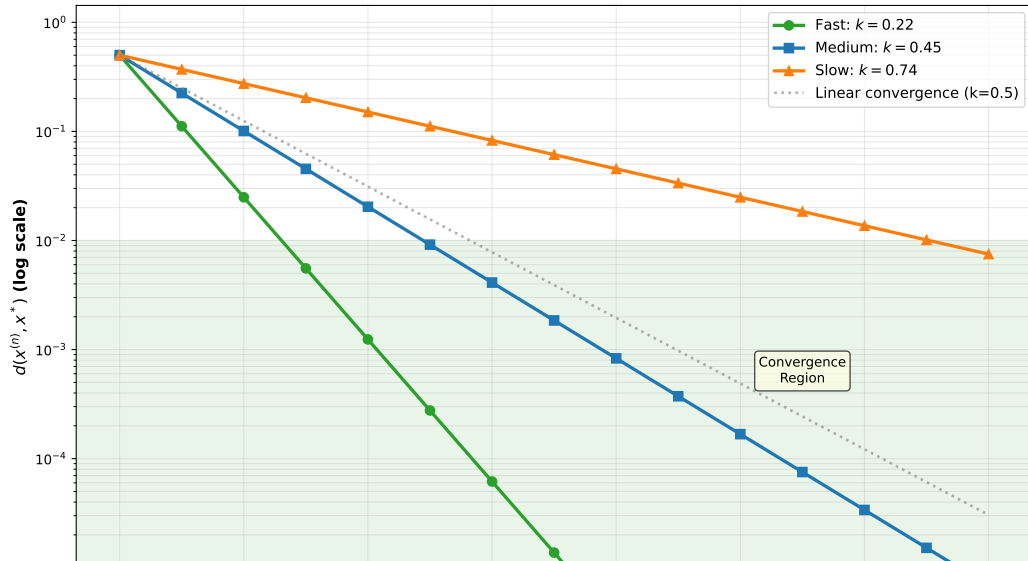
- 1 T has a unique fixed point p
- 2 For any $x_0 \in X$, the Picard iteration $x_{n+1} \in T(x_n)$ converges to p
- 3 The convergence is exponential

Comparison with Classical Nadler

- Nadler: constant contraction $H(T(x), T(y)) \leq k \cdot d(x, y)$
- Asymptotic: varying contraction with summable error $H(T^n(x), T^n(y)) \leq k_n \cdot d(x, y) + \alpha_n$
- More flexible for non-stationary iterations

Convergence Visualization

Convergence of Iterative Schemes for Multivalued Mappings



Applications in Differential Inclusions

Definition

A **differential inclusion** is a generalization of ODEs:

$$\dot{x}(t) \in F(t, x(t))$$

where F is a multivalued mapping.

Fixed Point Formulation: Using integral formulation:

$$x(t) = x_0 + \int_0^t f(s, x(s)) ds, \quad f(s, x) \in F(s, x)$$

Define integral operator T on function space:

$$T(x)(t) = x_0 + \int_0^t F(s, x(s)) ds$$

Then solutions are fixed points of T .

Existence Theorem

Applications in Game Theory

Non-Cooperative Games

Consider strategic game with players choosing from sets S_i (strategy multivalued mappings). Player payoff: $u_i(s_1, \dots, s_n)$.

Best Response Correspondence:

$$BR_i(s_{-i}) = \arg \max_{s_i \in S_i} u_i(s_i, s_{-i})$$

This is multivalued when multiple strategies maximize utility.

Theorem (Nash Equilibrium Existence)

If BR_i are u.s.c. with compact convex values, then by Kakutani's/Fan's theorem, there exists Nash equilibrium $(s_1^, \dots, s_n^*)$ such that $s_i^* \in BR_i(s_{-i}^*)$ for all i .*

Applications in Variational Inequalities

Problem

Find $x^* \in K$ such that

$$\langle Ax^*, y - x^* \rangle \geq 0 \quad \text{for all } y \in K$$

where K is closed convex and A is possibly multivalued operator.

Multivalued Setting:

$$0 \in T(x^*) + N_K(x^*)$$

where $N_K(x)$ is the normal cone at x in K .

Fixed Point Reduction

Using resolvent operator: $J_\lambda = (I + \lambda T)^{-1}$, solutions to VI correspond to fixed points of J_λ . This is again a multivalued fixed point problem.

Example: Control Systems

Problem

Linear control system:

$$\dot{x}(t) = Ax(t) + Bu(t), \quad u(t) \in U$$

where U is a compact constraint set.

Reachable Set: Define $R(t)$ = set of all states reachable at time t from initial state x_0 .

Can be characterized as:

$$R(t) = \bigcup_{u \in \mathcal{U}} \left\{ e^{At} x_0 + \int_0^t e^{A(t-s)} B u(s) ds \right\}$$

Fixed Point Perspective

In phase space, the reachable set evolves under multivalued "flow" map. Fixed point theorems for multivalued mappings can ensure:

- Existence of invariant sets
- Periodic orbits in control systems

Generalizations of Nadler's Theorem

Theorem (Assad and Kirk, 1976)

Let (X, d) be a complete metric space and $T : X \rightarrow CB(X)$. If for some $\phi : [0, \infty) \rightarrow [0, \infty)$ with $\phi(t) < t$ for $t > 0$:

$$H(T(x), T(y)) \leq \phi(d(x, y)) \quad \text{for all } x, y \in X$$

then T has a fixed point.

Other Extensions

- **Markin:** Weakly contractive multifunctions in Hilbert spaces
- **Smithson:** Contractive multifunctions with point-wise contractive values
- **Reich:** Set-valued nonexpansive mappings
- **Rhoades:** Various types of weak contractions for multivalued maps

Extension to b-Metric and Partial Metric Spaces

b-Metric Spaces

Metric d satisfying triangle inequality with constant $s \geq 1$:

$$d(x, z) \leq s[d(x, y) + d(y, z)]$$

Theorem

Fixed point theorems extend to b-metric spaces with Hausdorff metric appropriately modified.

Partial Metric Spaces

General metric allowing $d(x, x) > 0$ (non-identical elements at distance 0).

Current Research:

- Multivalued mappings in metric-like spaces
- Fuzzy-valued mappings
- Mappings with non-convex values

Open Problems and Future Directions

- **Uniqueness of Fixed Points:** Under what conditions is fixed point unique?
- **Stability Under Perturbations:** How robust are fixed points to small changes in T ?
- **Global Convergence:** Can we guarantee convergence from any starting point without local assumptions?
- **Computational Methods:** Efficient algorithms for approximating fixed points
- **Multivalued Versions:** Fixed point theory for other problem types (eigenvalue problems, optimization)
- **Applications:** New applications in machine learning, data analysis, uncertainty quantification

Main Results Summary

Theorem	Conditions	Conclusion
Fan (1952)	u.s.c., closed convex values, compact domain	Fixed point in K
Himmelberg (1972)	u.s.c., closed convex, compact range	Fixed point exists
Nadler (1969)	$H(T(x), T(y)) \leq k \cdot d(x, y)$ $k < 1$ on complete space	Unique fixed point
Asymptotic Contraction	$H(T^n(x), T^n(y))$ $\leq k_n d(x, y) + \alpha_n$ $\sum k_n, \sum \alpha_n < \infty$	Unique fixed point, exponential convergence

Key Concepts

- **Multivalued Mapping:** $T : X \rightarrow 2^X$ maps points to sets
- **Fixed Point:** $x^* \in T(x^*)$ (set membership, not equality)
- **Hausdorff Metric:** $H(A, B)$ measures distance between sets
- **Upper Semicontinuity:** Natural continuity notion for multifunctions
- **Contraction:** $H(T(x), T(y)) \leq k \cdot d(x, y)$ for $k < 1$
- **Topological Methods:** Compactness + convexity + continuity guarantee fixed point
- **Metric Methods:** Contractions guarantee unique fixed point with convergence

Connections to Other Topics

- **Differential Equations:** Solutions via integral reformulation as fixed points
- **Optimization:** Variational inequalities reduce to fixed point problems
- **Game Theory:** Nash equilibria are fixed points of best response mappings
- **Economics:** Market equilibrium, general equilibrium theory
- **Control Theory:** Reachable sets, invariant sets, stabilization
- **Numerical Analysis:** Iteration schemes, convergence analysis
- **Topology:** Degree theory, index theory, homotopy

Resources and Further Reading

Key References

- Pathak, H. K. (2018). *Introduction to Nonlinear Analysis and Fixed Point Theory*. Springer.
- Aubin, J. P., & Frankowska, H. (2009). *Set-Valued Analysis*. Birkhäuser.
- Denkowski, Z., Migórski, S., & Papageorgiou, N. S. (2003). *An Introduction to Nonlinear Analysis: Theory*. Springer.

Topics for Deeper Study

- Measure of noncompactness and condensing multivalued mappings
- Fixed points for mappings with non-convex values
- Multivalued versions of Mann and Ishikawa iterations
- Differential inclusions and their solutions

Conclusion I

- Multivalued mappings extend classical fixed point theory to set-valued functions
- The Hausdorff-Pompeu metric provides the natural framework
- Two main approaches:
 - **Topological:** Compactness + continuity + convexity (Fan, Himmelberg)
 - **Metric:** Contractions with complete space (Nadler)
- Rich applications in:
 - Differential inclusions
 - Variational inequalities
 - Game theory and economics
 - Control theory
- Active area of research with numerous generalizations and extensions
- Bridges pure mathematics with practical applications

Thank You!